How would you describe the "un-" prefix used in English, such as in "unnatural" or "unwarranted"?

The prefix "un-" is a common morpheme in English, attached to adjectives, participles, and sometimes nouns and verbs, to denote the negative, the opposite, the reverse, or the absence of the base word. Essentially, it's a way of forming antonyms or expressing negation.

For example:

- "Natural" means existing in or caused by nature; not made or caused by humankind. Adding "un-" creates "unnatural," meaning not in accordance with nature; artificial or not normally expected.
- "Warranted" implies something is justified or necessary. When you attach "un-" to form "unwarranted," it means something is not justified or called for under the circumstances.

The "un-" prefix is of Old English origin, and it is widely used to create new words that convey the sense of not being, not having, or not doing the concept of the root word. It's a productive prefix, meaning it is still actively used to create new terms.

I'm trying to decide how to name a few states in my code. There is some data that can take a "normalized" form and a "not normalized" form. For the latter I'd like to find a single word name. Two candidates came up: "non-normalized" and "unnormalized." While "non-normalized" is a bit harder on the tongue, I feel it's correct. I'm not sure about "unnormalized" -- it feels like "un" means taking away something, but in our case "not normalized" is the "natural" state of the data, and normalization gets applied on top of that, sort of. To me using "normalized" vs "unnormalized" would suggest the opposite relationship between these states, i.e. that being "normalized" is the normal thing, and somehow "unnormalized" is the more complicated, less natural state...

Not sure if this makes sense. Would you use "non-normalized", "unnormalized" or something else?

Your understanding of the connotations of "un-" is quite astute. The "un-" prefix can indeed sometimes imply a reversal or removal of a state, as in "undo" (to reverse an action) or "untie" (to reverse the state of being tied). However, in many cases, "un-" simply indicates the absence of a property without the implication that this is a reversal or that the property should have been present in the first place, such as "unhappy" or "uncertain."

For your specific case, both "non-normalized" and "unnormalized" could be suitable, but they might carry slightly different implications:

- "Non-normalized" is more neutral and simply states the absence of normalization without implying that normalization should have occurred. It suggests that the data has not been subjected to the process of normalization, and it's more of a passive state. It's commonly used in technical and scientific contexts where precision is valued.
- "Unnormalized" could suggest that the data was normalized at some point and has been changed back to its original state, or it could also just indicate the lack of normalization. However, in practice, "unnormalized" is often used in mathematics and computing to simply mean "not normalized," without necessarily implying a process of denormalization.

Given that "normalized" data implies that an active process of normalization has been applied, using "non-normalized" would unambiguously indicate that this process has not been applied, and there is no suggestion that it is a state that has been reverted to. Thus, "non-normalized" might be the more technically appropriate term for your case, emphasizing that this is the default or natural state.

But in the end, the choice may also be guided by convention within your domain, consistency with existing terminology in your codebase, or even the preference for brevity. If "unnormalized" is commonly used in your field and understood by your peers to mean "not normalized" without any additional implications, it could be perfectly acceptable to use that term for simplicity.